

Literature review of two existing exoplanet tools/apps

chose:

Planetary Habitability Laboratory Habitable Worlds Catalog (HWC) a public, habitability-focused catalog.

ExoMiner — a NASA/MAST machine-learning system (deep neural network) for vetting/validating transit signals (exoplanet candidate classifier). It's a research/operational ML tool that has been used to validate new planet.

PHL – Habitable Worlds Catalog (HWC)

NASA – ExoMiner

Our Project – AI-Powered Exoplanet Habitability Explorer

<u>Aspect</u>	<u>PHL - Habitable Worlds Catalog (HWC)</u>	<u>NASA -ExoMiner</u>	<u>Proposed Project – AI-Powered Exoplanet Habitability Explorer</u>
1. Objective	Rank and classify known exoplanets by habitability.	Validate exoplanet candidates using deep learning.	Predict and visualize exoplanet habitability using AI models.
2. Focus Area	Habitability assessment of confirmed planets.	Detection and confirmation of new planets.	Combined detection validation and habitability prediction.
3. Technology Used	Analytical formulas and empirical indices.	Deep Neural Network (Explainable AI).	Hybrid AI model integrating ML, data analytics, and visualization.
4. Data Source	Published exoplanet data (mass, radius, temperature).	Kepler and TESS light curve data.	Multiple sources: NASA Archive, HWC, and new datasets.
5. Output	Habitability score and classification list.	Probability of being a true planet.	Habitability probability, confidence level, and graphical explorer.

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6. User Interface	Public web-based catalog (static).	Research pipeline for NASA scientists (no public UI).	Interactive and user-friendly web dashboard.
7. Innovation Level	Limited AI use, focuses on cataloging.	Advanced AI for planet detection only.	Novel integration of AI-based validation and habitability prediction in one platform.

Summary

- **PHL (HWC):** Focus on habitability scoring of known planets.
- **ExoMiner:** Focus on validation of exoplanet candidates.
- **Our Project:** Combines both AI-powered detection and habitability prediction with visualization.